
PhysDNet: Physics-Driven Gradient Amplification for Real-Time Image Dehazing

Rayen Dhahri¹ Lei Sun² Victor Javier Kartsch Morinigo³
Alice Pagnoux³ Luca Benini^{3,4} Danda Pani Paudel² Luc Van Gool²

¹Independent Researcher

²INSAIT, Sofia University “St. Kliment Ohridski”

³ETH Zurich

⁴Department of Electrical, Electronic and Information Engineering (DEI), University of Bologna

Abstract

Modern dehazing networks are often computationally expensive and overfit to synthetic haze, limiting real-time deployment. This underscores an urgent demand for efficient, high-fidelity restoration methods that can operate on resource-constrained edge devices in practical, dynamic environments. We introduce PhysDNet, a lightweight multimodal architecture that fuses NIR imagery with sparse LiDAR depth and is trained with a physics-constrained dual-head objective. Differentiating through a fixed Koschmieder inversion induces *density-adaptive gradient amplification*, acting as an implicit resource allocation mechanism that concentrates learning on heavily degraded regions without increasing inference cost. PhysDNet delivers superior restoration with minimal capacity and enhanced training efficiency, facilitating real-time deployment on edge devices without increasing inference overhead. On the Seeing Through Fog benchmark, PhysDNet outperforms competitors by **1.67 dB** at **5.2×** lower compute and recovering **54.9%** of fog-induced detection loss. Ablations and cross-domain evaluations show that physics-guided training improves generalization including under non-homogeneous haze that violates the Koschmieder assumption. PhysDNet further demonstrates compatibility with fixed-function NPUs, running at **29/45/99 FPS** at **<4 W** across L/M/S variants. Anonymous code, deployment scripts and pre-trained checkpoints are available at <https://anonymous.4open.science/r/PhysDNet-1D62/>.

1 Introduction

Image dehazing is foundational for outdoor perception under adverse weather, particularly autonomous driving where reduced visibility compromises detection and tracking [1]. Modern autonomous platforms already carry the sensors needed: time-gated NIR cameras that reject nearby backscatter, and LiDAR providing weather-invariant depth [2]. Active time-gated NIR provides a hardware-level bypass to RGB’s failure under dense scattering: synchronized exposure with emitted laser pulses rejects atmospheric backscatter and recovers structural semantics where RGB is effectively blind. Yet most learned dehazing methods [3–7] process RGB alone and chase quality by scaling capacity, deeper backbones, wider attention, iterative refinement, inflating inference cost and relying on operators such as self-attention and layer normalization, which are incompatible with fixed-function edge accelerators [8]. Consequently, a practical dehazing solution must be lightweight enough for deployment on low-power edge accelerators while seamlessly exploiting the multi-modal sensor suites already ubiquitous on autonomous platforms.

We introduce **PhysDNet**, a lightweight multimodal dehazing network for real-time edge deployment. PhysDNet fuses gated NIR imagery with sparse LiDAR depth through mask-aware multi-scale gating

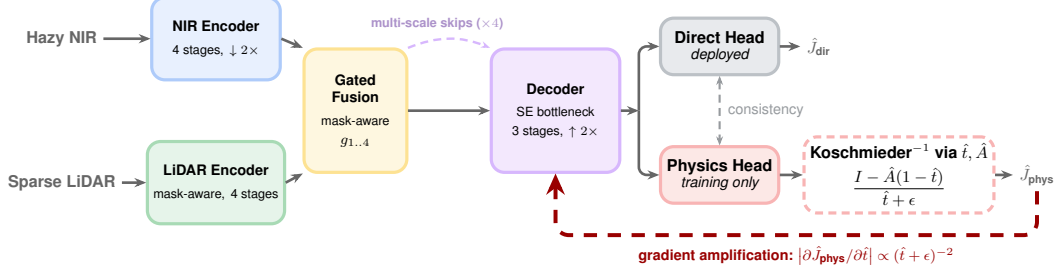


Figure 1: **PhysDNet architecture.** Dual encoders extract NIR and sparse-depth features, merged by mask-aware gating with Squeeze-and-Excitation at the bottleneck. The shared representation feeds a direct head (deployed at inference) and a physics head (training only). Backpropagating through the Koschmieder inversion scales gradients by $(\hat{t} + \epsilon)^{-2}$, concentrating optimization on heavily degraded regions at zero inference cost.

and is trained with a *physics-constrained dual-head framework*: a deployable direct restoration head and a parameter-free auxiliary head that inverts the Koschmieder scattering model [9].

Our central insight is that the auxiliary physics head improves learning not by constraining the final output, but by shaping the *optimization dynamics*. Backpropagating through the fixed physical inversion induces transmission-dependent gradient scaling proportional to $(t + \epsilon)^{-2}$, automatically amplifying supervision in heavily degraded regions. We call this *density-adaptive gradient amplification*: a hyperparameter-free implicit curriculum that prioritizes the hardest pixels during training at zero inference cost. **To our knowledge, this is the first analysis of differentiation through a physical inversion as a mechanism for shaping optimization via structured gradient scaling.** This mechanism improves performance through more efficient use of gradients during training rather than increased model capacity. In contrast to scaling architectures, which increases inference cost, PhysDNet achieves its gains by reallocating optimization focus, leaving the deployment footprint unchanged.

We deliver the following contributions:

- **Density-adaptive gradient amplification.** A dual-head training framework where a fixed inversion induces $(t + \epsilon)^{-2}$ gradient scaling, focusing learning on degraded regions without explicit weighting or extra inference cost.
- **State-of-the-art accuracy at low compute.** On STF, PhysDNet reaches 36.46 dB PSNR at 18.98 GMACs, outperforming FFA-Net by +1.67 dB at $5.2\times$ lower compute, and recovers 54.9% of YOLOv8 detection loss under aggressive fog.
- **Edge deployability.** The only evaluated method combining competitive restoration quality with fixed-function NPU compatibility, running at 29-99 FPS at <4 W under standard INT8 quantization.

2 Method

We adopt the Koschmieder scattering model [9]: for a pixel x , the hazy intensity $I(x)$ is a convex combination of scene radiance $J(x)$ and global airlight A ,

$$I(x) = J(x)t(x) + A(1 - t(x)), \quad t(x) = e^{-\beta d(x)}, \quad (1)$$

with transmission $t(x) \in (0, 1]$, scattering coefficient β , and depth $d(x)$. Single-image recovery is ill-posed in both t and A [10]; PhysDNet breaks this ambiguity by fusing sparse LiDAR depth (weather-invariant) with time-gated NIR imagery (rejects nearby backscatter) [2]. Extended background in Appendix B.

We present **PhysDNet (Physics Dual-head Dehazing Network)**, a lightweight architecture for LiDAR-guided image dehazing. A fixed, non-learnable physics inversion head operates in parallel with a data-driven restoration head, both sharing a common encoder-decoder backbone. Backpropagation through the physics inversion induces $(\hat{t} + \epsilon)^{-2}$ gradient scaling, concentrating optimization on severely degraded regions without additional parameters.

The model receives as input a hazy image $I \in [0, 1]^{3 \times H \times W}$, sparse LiDAR depth $D_s \in \mathbb{R}^{1 \times H \times W}$, and a binary validity mask $M \in \{0, 1\}^{1 \times H \times W}$, and jointly predicts a restored image, a per-pixel transmission map, and a global airlight vector.

2.1 Network Architecture

PhysDNet is a dual-branch encoder-decoder (fig. 1) with gated cross-modal fusion and two output heads. Two modality-specific encoder branches process NIR and depth independently at four spatial scales, merging features at each scale via a learned gate. A single decoder upsamples the fused representations. Unlike a standard U-Net [11], each branch learns modality-specific representations before fusion, with gating adapting to local signal availability.

Dual encoder and decoder. Two modality-specific encoder branches follow a four-stage downsampling structure using depthwise-separable (DWSep) convolutions [12], BatchNorm, and ReLU6 [13], with residual shortcuts and $2 \times$ max-pooling per stage for edge efficiency. The image branch accepts 3-channel NIR input; the depth branch receives the sparse depth D_s concatenated with the validity mask M . A Squeeze-and-Excitation (SE) channel attention block [14] is applied once at the bottleneck after gated fusion. The decoder applies three bilinear upsampling stages with skip connections from each encoder scale. Full layer-wise specifications are in Appendix D.

Gated fusion. At each of the four encoder scales, the image features $\mathbf{f}_{\text{img}}$ and depth features $\mathbf{f}_{\text{depth}}$ are combined via a learned gate:

$$g = \sigma(W_g[\mathbf{f}_{\text{img}}; \mathbf{f}_{\text{depth}}]), \quad \mathbf{f}_{\text{fused}} = g \odot \mathbf{f}_{\text{img}} + (1 - g) \odot \mathbf{f}_{\text{depth}}, \quad (2)$$

where $[\cdot; \cdot]$ denotes channel-wise concatenation and W_g is a 1×1 convolution following the gated multimodal fusion paradigm [15, 16]. This mechanism is inherently mask-aware: where LiDAR returns are absent ($M=0$), the depth features carry no geometric signal and the gate learns to rely on the image branch; where depth is present, geometric cues help disambiguate haze density from scene content.

Model family. Scaling the base channel width $c \in \{32, 64, 96\}$ yields three variants (S/M/L) spanning the accuracy, efficiency Pareto front (table 10). Efficiency metrics are presented in table 1; even PhysDNet-L (18.98 GMACs) remains lower-cost than evaluated baselines.

2.2 Physics-Constrained Dual-Head Design

The shared decoder features $\mathbf{z} \in \mathbb{R}^{c \times H \times W}$ feed two complementary output heads. This dual-head design is the core mechanism through which physical knowledge is injected into the learning process.

Head 1: Direct restoration (data-driven). A convolutional projection with sigmoid activation maps decoder features directly to the image domain, yielding $\hat{J}_{\text{dir}}(x) \in [0, 1]^3$. This unconstrained estimator makes no assumption about the physical image formation process and serves as the deployed output, providing robustness against Koschmieder-model violations (spatially varying airlight, multiple scattering, sensor artifacts).

Head 2: Physics-guided restoration (structured). Two auxiliary sub-heads predict the atmospheric scattering parameters from the same shared features:

$$\hat{t}(x) = t_{\min} + (t_{\max} - t_{\min}) \sigma(W_t * \mathbf{z}(x) + b_t) \in [t_{\min}, t_{\max}], \quad (3)$$

$$\hat{A} = \sigma(W_A \cdot \text{GAP}(\mathbf{z}_{\text{bot}}) + b_A) \in [0, 1]^3, \quad (4)$$

where W_t is a 1×1 convolution producing per-pixel transmission, W_A is a fully connected layer applied after global average pooling of the bottleneck features $\mathbf{z}_{\text{bot}}$, and $t_{\min}=0.05$, $t_{\max}=0.95$ prevent numerical instability.

The physics-restored image is obtained by inverting Eq. (1) via its closed-form solution:

$$\hat{J}_{\text{phys}}(x) = \text{clamp}\left(\frac{I(x) - \hat{A}(1 - \hat{t}(x))}{\hat{t}(x) + \epsilon}, 0, 1\right), \quad (5)$$

with $\epsilon = 10^{-4}$. This inversion contains **zero learnable parameters**: it is a fixed, differentiable function whose output quality is determined solely by the accuracy of $\hat{t}$ and $\hat{A}$. During training,

gradients flow backwards through this fixed computation into the shared encoder, producing the structured gradient signal analysed in §2.3.

Inference behaviour. At test time only the direct head is deployed; the physics head operates solely during training. This ensures reported gains reflect improved features rather than test-time physics exploitation.

2.3 Density-Adaptive Gradient Amplification

We now derive the gradient signal the physics head propagates into the shared encoder. The structure reveals an implicit curriculum that emerges from the scattering inversion itself, without explicit sample weighting or scheduling.

Taking partial derivatives of Eq. (5) (within the valid range, before clamping) with respect to the predicted scattering parameters:

$$\frac{\partial \hat{J}_{\text{phys}}}{\partial \hat{t}} = \frac{\hat{A} - I(x)}{(\hat{t} + \epsilon)^2}, \quad (6) \quad \frac{\partial \hat{J}_{\text{phys}}}{\partial \hat{A}} = \frac{\hat{t} - 1}{\hat{t} + \epsilon}, \quad (7)$$

where we neglect the $O(\epsilon)$ term in the numerator of Eq. (6). In practice, $\epsilon = 10^{-4}$ remains several orders of magnitude smaller than $t_{\min} = 0.05$, ensuring the approximation holds even in the lowest-transmission regime.

Transmission gradient: implicit hard-example mining. The factor $(\hat{t} + \epsilon)^{-2}$ in Eq. (6) implies that gradient magnitude scales *inversely with the square of transmission*. In dense fog ($\hat{t} \rightarrow t_{\min} = 0.05$), gradients are amplified by up to $(t_{\min} + \epsilon)^{-2} \approx 400\times$ relative to clear-air pixels ($\hat{t} \approx 1$), concentrating learning on heavily degraded regions. This scaling is empirically confirmed in fig. 3(a): PhysDNet-M follows the $(\hat{t} + \epsilon)^{-2}$ curve and exceeds the non-physics baseline by up to $\sim 694\times$ in dense haze (fig. 3(b)).

This behaviour acts as an implicit resource allocation mechanism during training, analogous to hard-example mining [17], where degraded pixels dominate the gradient signal. Unlike explicit weighting schemes (e.g., focal loss [18]), this effect requires *no additional hyperparameters* and arises directly from differentiating through the physics inversion.

Variance correspondence. The same t^{-2} structure governs the inversion’s noise variance, $\text{Var}[\hat{J}_{\text{phys}}] = \sigma^2/t^2 + O(\delta)$ (Appendix F, Prop. 1), making the physics head the structural inverse of classical heteroscedastic regression [19], which *down-weights* high-variance samples. Full derivations, airlight-gradient analysis, bounded-gradient discussion, and per-pixel spatial validation on real STF fog are in Appendix F.

2.4 Physics-Anchored Residual Learning

The physics-reconstruction loss $\mathcal{L}_{\text{pr}}$ anchors the shared latent representation in a physically meaningful space, while the standard reconstruction loss $\mathcal{L}_{\text{rec}}$ lets the direct head learn residual corrections for non-homogeneous fog and sensor effects beyond Koschmieder, ensuring the network is physics-anchored but not physics-limited. A physics-aware contrastive loss $\mathcal{L}_{\text{contra}}$ uses physics-head artifacts as dynamic hard negatives. Full multi-term objective and late-activation schedule in Appendix E.

3 Experiments

We evaluate PhysDNet along three axes. First, **quality** on STF (§3.2–§3.3), out-of-distribution fog (§3.4), and real STF fog (§3.7). Second, **utility** for downstream detection (§3.5) and zero-shot cross-domain transfer (§3.6). Third, **deployability** on two automotive edge platforms (§3.8). Appendices report LiDAR-sparsity robustness (App. N) and no-reference perceptual quality (App. O).

3.1 Experimental Setup

We train and evaluate on the Seeing Through Fog (STF) dataset [1], using 4 212 training frames and 1 299 validation frames with on-the-fly synthetic haze generated via Eq. (1). Sparse LiDAR is used unmodified at both train and test time. Dataset details in App. P.

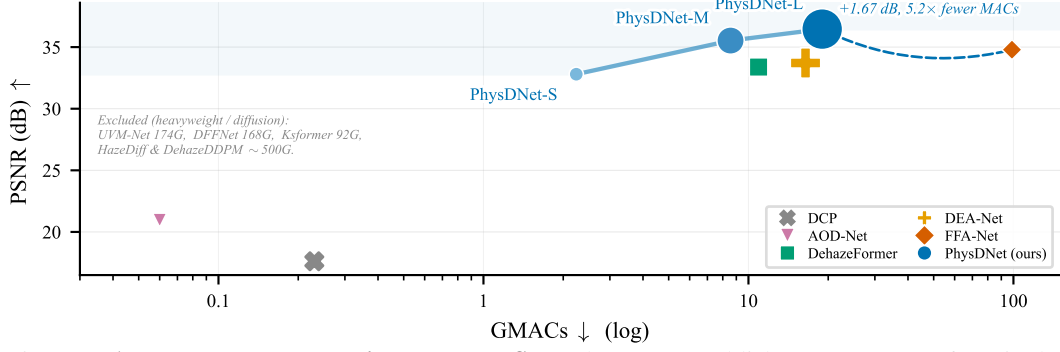


Figure 2: **Accuracy vs. computational cost on STF.** PhysDNet establishes a new Pareto front in the edge-deployable region. Baselines are plotted at their best-performing resolution (Table 1). PhysDNet-L outperforms FFA-Net by +1.67 dB at $5.2\times$ lower compute; PhysDNet-S retains competitive accuracy at ultra-low compute. Aside from PhysDNet, AOD-Net compiles to the NPU but its restoration quality is insufficient for deployment. Heavyweight and diffusion baselines excluded (App. C).

We compare against Dark Channel Prior (DCP) [20], AOD-Net [3], FFA-Net [4], DehazeFormer-T [5], and DEA-Net [6]. All PhysDNet variants are trained with AdamW at 256×256 using only NPU/TensorRT-friendly operators (depthwise-separable convolutions, BatchNorm, ReLU6, bilinear upsampling). All reported results correspond to the unconstrained direct head, ensuring gains cannot stem from trivial consistency with the synthetic generation process. Recent heavyweight (UVM-Net, DFFNet, Ksformer) and diffusion-based (HazeDiff, DehazeDDPM, FrDiff) methods are excluded as they exceed real-time edge constraints; selection rationale in App. C. Training details in App. E.

3.2 Restoration Quality on STF

Table 1 compares PhysDNet against four learned baselines and the classical DCP [20]. At matched 256×256 resolution, PhysDNet-L reaches 36.46 dB PSNR, outperforming FFA-Net+LiDAR at its preferred 512×512 resolution (34.79 dB) by +1.67 dB at $5.2\times$ lower compute. PhysDNet-M (2.95 M, 8.56 GMACs) surpasses DEA-Net by +1.82 dB at roughly half the cost. The gap persists across resolutions, and §3.3 confirms the physics-constrained training regime, not backbone capacity, is the primary driver. Qualitative comparisons in Appendix Q.

3.3 Physics Head Ablation

To isolate the effect of physics-constrained training, we remove the physics head (*NoPhy*) from an identical ch64 model, disabling transmission/airlight prediction and all physics losses. This reduces parameters by only 0.04 M ($2.95\rightarrow 2.91$ M) but causes a large drop in performance (Table 2).

Table 2: **Physics head ablation.** Removing the physics head degrades PSNR by 5.51 dB and reduces downstream mAP recovery.

Variant	Physics Head	Params (M)	PSNR (dB)	SSIM	mAP (%)	Δ mAP
PhysDNet-M	✓	2.95	35.55	0.9569	11.70	+3.05
PhysDNet-M NoPhy	×	2.91	30.02	0.9263	10.70	+2.06
<i>Difference</i>		−0.04	−5.51	−0.0306	−1.00	−0.99

Removing the physics head reduces PSNR by **5.51 dB** and mAP recovery from +3.05 to +2.06, indicating that gains arise from the training regime rather than architecture alone.

Gradient amplification. Figure 3 confirms the $(\hat{t} + \epsilon)^{-2}$ scaling: PhysDNet-M gradients follow the theoretical curve and exceed NoPhy by up to $\sim 694\times$ in dense haze, reflecting implicit hard-example mining.

Table 1: **Quantitative comparison on the STF validation set.** PSNR/SSIM for the *direct* restoration head. PhysDNet-M reports mean $\pm$ SE over 3 seeds. Best in **bold**, second underlined. GMACs at the indicated resolution.

Method	Input	Res.	Params (M)	GMACs	PSNR (dB) $\uparrow$	SSIM $\uparrow$
DCP [20]	NIR	256	0	—	17.62	0.5992
AOD-Net [3]	NIR	512	0.002	0.23	18.62	0.7123
AOD-Net	NIR	256	0.002	0.06	20.69	0.7499
AOD-Net	LiDAR+Gated	512	0.002	0.24	17.05	0.6669
AOD-Net	LiDAR+Gated	256	0.002	0.06	20.95	0.7329
DehazeFormer-T [5]	NIR	512	1.36	10.92	33.38	0.9348
DehazeFormer-T	NIR	256	1.36	2.73	32.22	0.9398
DehazeFormer-T	NIR+LiDAR	512	1.36	10.97	29.91	0.9092
FFA-Net [4]	NIR	512	0.77	98.57	33.90	0.9474
FFA-Net	NIR	256	0.77	24.64	33.80	0.9504
FFA-Net	NIR+LiDAR	512	0.77	98.73	34.79	0.9516
FFA-Net	NIR+LiDAR	256	0.77	24.68	33.23	0.9476
DEA-Net [6]	NIR	256	3.65	16.42	33.71	0.9588
DEA-Net	LiDAR	256	3.65	16.44	33.53	0.9572
PhysDNet-S	NIR+LiDAR	256	0.72	2.24	32.80	0.9331
PhysDNet-M	NIR+LiDAR	256	2.95	8.56	<u>35.55$\pm$0.07</u>	0.9566 $\pm$ 0.0009
PhysDNet-L	NIR+LiDAR	256	6.59	18.98	36.46	0.9606

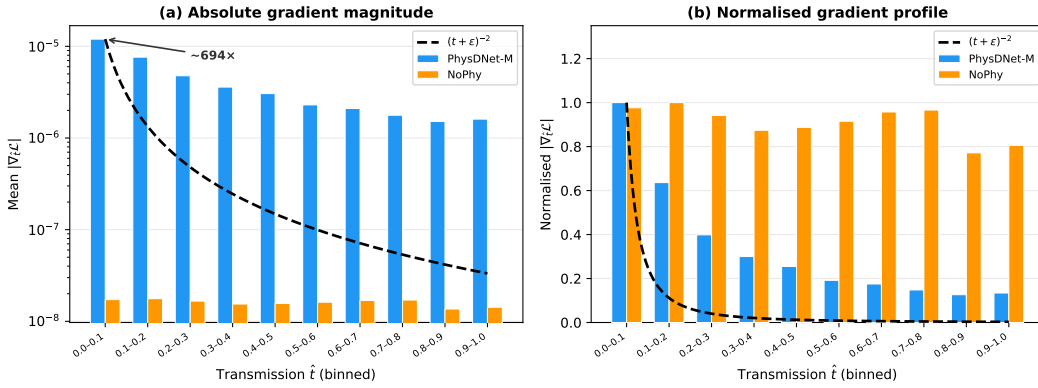


Figure 3: **Density-adaptive gradient amplification.** PhysDNet-M follows the $(t + \epsilon)^{-2}$ scaling and concentrates gradients on low-transmission regions, unlike the flat NoPhy baseline.

3.4 Out-of-Distribution Fog Robustness

We sweep $\beta \in [0.005, 0.10]$ at $A=0.8$ on the full validation set (1 299 frames per β), with no retraining (see Appendix R for the full performance curve); values $\beta \geq 0.05$ lie outside the training range. PhysDNet-M maintains the highest PSNR across the entire range and degrades smoothly beyond the training boundary, retaining a >6 dB advantage over FFA-Net at $\beta=0.06$. We attribute this robustness to the physics-constrained encoder: because the inversion formula (Eq. 5) is domain-invariant, the learned (t, A, J) factorisation generalises to unseen β values without retraining.

3.5 Downstream Object Detection

Pixel-level metrics on synthetic data do not guarantee that semantically meaningful structure has been recovered. We therefore measure the impact of dehazing on downstream object detection using YOLOv8-M [21] (pretrained on COCO, $\text{conf} \geq 0.25$, $\text{IoU} \geq 0.5$) on STF validation frames with synthetically added *aggressive* fog ($\beta=0.06$, out-of-distribution). Full protocol details are in section K. We report relative recovery rather than absolute performance, as the detector is not fine-tuned for STF.

PhysDNet-L recovers 3.12 pp (**54.9%** of the degradation) and PhysDNet-M 3.05 pp (53.7%), while the best modern baseline, DEA-Net (NIR), recovers only 2.45 pp (43.1%). This indicates that

Table 3: **Downstream object detection** on 426 STF val frames with synthetic fog ($\beta=0.06$). Detector: YOLOv8-M ($\text{conf} \geq 0.25$, $\text{IoU} \geq 0.5$). ΔmAP is relative to the foggy baseline; recovery = $\Delta\text{mAP}/|\text{fog degradation}|$.

Condition	AP_{veh}	AP_{pers}	AP_{eye}	mAP (%)	ΔmAP
Clean (upper bound)	33.95	7.97	1.05	14.33	—
Foggy ($\beta=0.06$)	18.09	5.42	2.42	8.64	-5.68
AOD-Net (NIR)	15.14	4.66	1.27	7.02	-1.62
AOD-Net (LiDAR+Gated)	14.40	4.23	1.49	6.70	-1.94
DehazeFormer-T (NIR)	16.37	5.74	1.65	7.92	-0.72
DehazeFormer-T (LiDAR)	15.62	5.24	1.21	7.36	-1.29
FFA-Net (NIR)	24.37	6.30	1.32	10.66	+2.02
FFA-Net (LiDAR)	23.09	5.35	1.39	9.94	+1.30
DEA-Net (NIR)	25.82	5.82	1.66	11.10	+2.45
DEA-Net (LiDAR)	25.52	5.72	2.00	11.08	+2.44
PhysDNet-S	24.49	5.94	1.22	10.55	+1.91
PhysDNet-M	27.65	6.45	0.99	11.70	+3.05
PhysDNet-L	27.60	6.70	0.99	11.76	+3.12

Recovery: PhysDNet-L=**54.9%**, PhysDNet-M=53.7%, DEA-Net (NIR)=43.1%, FFA-Net (NIR)=35.6%

Table 4: **Zero-shot cross-domain generalization** on O-HAZE and I-HAZE. All models trained on STF; no LiDAR at test time.

Method	O-HAZE [22] (outdoor)				I-HAZE [23] (indoor)			
	PSNR-Y $\uparrow$	SSIM-Y $\uparrow$	PSNR-RGB $\uparrow$	SSIM-RGB $\uparrow$	PSNR-Y $\uparrow$	SSIM-Y $\uparrow$	PSNR-RGB $\uparrow$	SSIM-RGB $\uparrow$
AOD-Net	15.11	0.5000	14.88	0.4931	14.69	0.5824	14.51	0.5724
FFA-Net	19.20	0.7177	18.49	0.6968	15.82	0.6521	15.48	0.6402
DehazeFormer-T	19.48	0.6839	18.81	0.6641	15.86	0.6142	15.56	0.6050
DEA-Net	19.92	0.7042	19.20	0.6854	16.31	0.6455	16.02	0.6362
PhysDNet-S	19.89	0.7190	19.21	0.6989	16.81	0.6895	16.49	0.6734
PhysDNet-M	20.45	0.7461	19.65	0.7253	16.93	0.6947	16.59	0.6795
PhysDNet-M (NoPhy)	18.92	0.7358	18.38	0.7174	16.33	0.6907	16.06	0.6746
PhysDNet-L	20.31	0.7488	19.51	0.7279	16.57	0.6887	16.24	0.6756

PhysDNet restores semantically relevant structure rather than only improving pixel-level metrics. A representative qualitative example is shown in fig. 4.

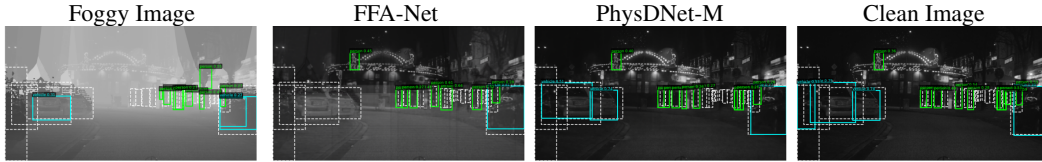


Figure 4: **Downstream detection after dehazing** ($\beta=0.06$). PhysDNet-M recovers scene contrast and enables more accurate detections compared to FFA-Net. Dashed white: ground truth; solid coloured: predictions.

3.6 Zero-Shot Cross-Domain Generalization and Non-Homogeneous Fog

We evaluate zero-shot on **O-HAZE** [22] and **I-HAZE** [23], imposing a quadruple shift (NIR $\rightarrow$ RGB, synthetic $\rightarrow$ real, LiDAR $\rightarrow$ none, outdoor $\rightarrow$ indoor). No fine-tuning. Due to near-achromatic outputs, we report luminance (Y) metrics following prior work. Diffusion-based methods are excluded as non-real-time.

PhysDNet remains competitive without depth input. PhysDNet-M achieves the best overall performance, improving over DEA-Net by +0.53 dB on O-HAZE and +0.62 dB on I-HAZE, while PhysDNet-S matches or exceeds all baselines on I-HAZE. The same advantage holds on NH-HAZE (non-homogeneous outdoor haze, App. M): PhysDNet-M outperforms DEA-Net by +0.15 dB PSNR-Y; removing the physics head reduces PSNR-Y by 0.54 dB, confirming that physics-guided training transfers despite the test distribution violating the homogeneous-scattering assumption.

Table 5: **Dual-head consistency on real STF fog.** High PSNR and correlation indicate strong agreement.

Metric	Dense Fog	Light Fog	Interpretation
MSE	0.0002	0.0006	Very low gap
PSNR (dB) $\uparrow$	37.48	33.83	High agreement
Pearson r $\uparrow$	0.9947	0.9912	Near-perfect match

Table 6: **Unified edge-deployment benchmark** (256×256 , batch 1, 50 warmup + 300 timed iterations on the Jetson Orin Nano). Fixed-function NPU uses INT8 post-training quantization; Jetson uses TensorRT FP16. $\times$ indicates compilation failure due to unsupported operators. Full CUDA FP32 numbers and power profiles in Appendix G.

Model	Params (M)	GMACs	Fixed-function NPU		Jetson Orin Nano	
			Compile	FPS	FPS	$\bar{P}$ (W)
AOD-Net LiDAR	0.002	0.06	$\checkmark$	1220	327.0	4.94
FFA-Net LiDAR	0.77	24.68	$\times$	—	46.9	8.21
DehazeFormer-T LiDAR	1.36	2.73	$\times$	—	47.1	12.38
DEA-Net LiDAR	3.65	16.44	$\times$	—	28.5	10.17
PhysDNet-S (ours)	0.72	2.24	$\checkmark$	99	108.3	13.45
PhysDNet-M (ours)	2.95	8.56	$\checkmark$	45	58.5	12.38
PhysDNet-L (ours)	6.59	18.98	$\checkmark$	29	38.8	11.80

The NoPhy ablation isolates the physics head: removing it reduces PSNR-Y by 1.53 dB on O-HAZE and 0.60 dB on I-HAZE. While smaller than the in-domain gap, this confirms that physics-guided training improves cross-domain generalization beyond backbone capacity.

Additional qualitative results and chrominance recovery are provided in App. L. Robustness to LiDAR sparsity is shown in App. N, where PhysDNet-M retains a +4.10 dB advantage at 0% depth.

3.7 Real-World Fog Validation

Since STF real-fog frames lack ground truth, we evaluate via **dual-head consistency**, i.e., agreement between $\hat{J}_{\text{dir}}$ and $\hat{J}_{\text{phys}}$ (Table 5).

Both heads show near-perfect agreement (PSNR > 30 dB, $r > 0.99$), confirming that the physics inversion produces outputs consistent with the direct head. Consistency is higher in dense fog (37.48 vs. 33.83 dB), aligning with the density-adaptive gradient amplification (§2.3).

We further validate real-haze performance using no-reference metrics (App. O): PhysDNet-M achieves the best NIQE on O-HAZE (3.92 vs. 4.64 for DEA-Net), confirming transfer beyond synthetic data.

3.8 Edge Deployment

PhysDNet’s restriction to NPU-friendly operators avoids softmax-attention, LayerNorm, and deformable convolutions, which block baseline compilation.

We validate deployability on two representative edge targets: (i) a fixed-function 26 TOPS NPU (< 4 W TDP), representative of strict automotive perception SoCs, and (ii) the NVIDIA Jetson Orin Nano Super (8 GB), representative of more permissive ADAS compute platforms. All models are exported to ONNX from the same trained checkpoints and evaluated at 256×256 . Table 6 reports compilation status, throughput, and power.

Fixed-function NPU. All models are compiled via vanilla INT8 post-training quantization (64-sample calibration, no QAT). The ~ 1.1 dB PSNR drop (App. G.1) is a conservative lower bound on quantization robustness. fig. 5 shows that within the training range, the INT8 output is visually indistinguishable from the FP32 reference.

Under our evaluation protocol (standard ONNX export and vendor compiler with INT8 post-training quantization), only PhysDNet and AOD-Net successfully compile, but AOD-Net’s restoration quality is insufficient for deployment (20.95 dB, §3.2). All other baselines fail due to

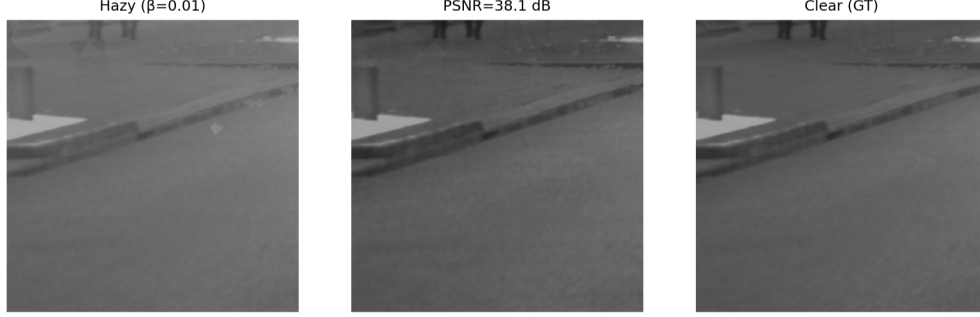


Figure 5: **INT8 NPU inference on a representative STF sample** ($\beta=0.01$). Hazy input, PhysDNet-M restored on the fixed-function NPU (38.1 dB), and ground truth. INT8 deployment preserves restoration quality within the training range; full β -sweep in App. G.1.

unsupported operators (e.g., attention, LayerNorm). PhysDNet is thus the only method that combines competitive accuracy with NPU compatibility, while exceeding the 10 Hz LiDAR rate.

Jetson Orin Nano. On the Jetson Orin Nano all candidates compile under CUDA+TensorRT. PhysDNet-M outperforms FFA-Net by $1.25\times$ despite $4\times$ more GMACs; PhysDNet-L outperforms DEA-Net by $2.05\times$. AOD-Net’s 327 FPS lacks deployable quality (§3.2).

4 Related Work

Early dehazing methods relied on hand-crafted priors such as the Dark Channel Prior [20], while later deep models learned direct restoration mappings using CNNs, attention, and transformers [3–6]. Recent physics-guided approaches embed the Koschmieder model as an output-consistency constraint [24, 25]; in contrast, PhysDNet uses physics to shape *gradient flow* during training rather than constrain outputs. Diffusion-based methods [26, 7] achieve high fidelity but require iterative sampling that precludes real-time use. Depth-aware dehazing has used binocular disparity, monocular depth, and dense LiDAR to constrain transmission [27–30], typically assuming dense geometric input; PhysDNet instead fuses highly sparse LiDAR directly with gated NIR via learned multi-scale gating, avoiding dense depth completion. By confining the physics inversion to training and restricting the backbone to edge-oriented operators [8], PhysDNet achieves zero inference overhead alongside its physical grounding.

5 Conclusion

We introduced PhysDNet, a lightweight multimodal dehazing architecture that uses a physics-constrained dual-head objective to shape optimization dynamics. Differentiating through a fixed Koschmieder inversion induces density-adaptive gradient amplification, concentrating learning on severely degraded regions at zero inference cost. This mechanism yields state-of-the-art performance on Seeing Through Fog, consistent gains under cross-domain evaluation, and robustness to LiDAR sparsity. PhysDNet is also the only evaluated method that combines strong restoration quality with fixed-function low-power NPU deployment, achieving 29-99 FPS at <4 W under INT8 quantization.

Beyond dehazing, our results suggest that physical models can serve not only as priors on outputs, but as mechanisms for shaping the learning signal itself, offering a new paradigm for efficient and deployable perception under physical constraints.

Limitations. PhysDNet is trained on synthetic Koschmieder haze; NoPhy ablations on three real-fog benchmarks (O-HAZE, I-HAZE, NH-HAZE) confirm transfer, including under non-homogeneous fog (App. M). Broader cross-platform multi-modal validation beyond STF remains open. Without color supervision, the model produces near-achromatic RGB outputs requiring heuristic chrominance recovery (§3.6). Extending the theoretical analysis beyond Koschmieder to other inverse problems with spatially-varying degradation is a natural next direction.

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A Broader impact

PhysDNet improves the ability of resource-constrained edge systems to perceive in adverse weather, with direct applications to autonomous driving, robotics, and ADAS. By recovering semantically meaningful structure and improving downstream object detection in dense fog, it may help reduce weather-related accidents and improve road safety. Its design also supports *Green AI* by enabling high-quality, physics-grounded restoration on ultra-low-power fixed-function NPUs ($< 4\text{ W}$), reducing reliance on energy-intensive GPU inference.

At the same time, this technology has dual-use potential and could support unconsented surveillance or military deployment in conditions where poor visibility would otherwise provide cover. It also introduces safety risks in critical systems if exposed to out-of-distribution weather, such as heterogeneous smoke or severe occlusions that violate the assumed Koschmieder model. For this reason, PhysDNet should be used only as part of a sensor-fusion stack, together with uncertainty estimation and hardware fail-safes.

B Extended Background

Ill-posed nature of monocular dehazing. Recovering $J(x)$ from a single $I(x)$ is fundamentally ill-posed: both $t(x)$ and A are unknown, and hand-crafted priors such as the Dark Channel Prior [10] frequently collapse under non-homogeneous real-world fog.

Active sensing and gated imaging mechanics. Modern autonomous platforms combine sparse LiDAR, which provides weather-invariant depth measurements $d(x)$, with active time-gated NIR imaging [2]. By delaying sensor exposure after emitting a synchronised NIR laser pulse, a gated camera physically rejects nearby backscatter and integrates only photons from a target depth slice, drastically improving the signal-to-noise ratio in dense fog. This gated imagery combined with sparse LiDAR forms the physical foundation of PhysDNet’s multi-modal input.

C Baseline Selection Rationale

Table 7: **Model Selection and Deployment Potential.** Single-image dehazing for real time perception is strictly bottlenecked by the latency ($< 33\text{ ms}$) and power ($\leq 15\text{ W}$) constraints of edge hardware. We therefore restrict our evaluation to single-pass architectures capable of running within these limits. Recent massive architectures and diffusion models are explicitly excluded, as their computational demands or iterative sampling mechanics inherently preclude real-time deployment.

Method	Venue	Compute (GFLOPs)	Edge Viable	Rationale for Inclusion / Exclusion
Evaluated Baselines (Real-Time / Single-Pass)				
DCP [20]	CVPR '09	—	✓	Zero-parameter physical prior anchor.
AOD-Net [3]	ICCV '17	0.06	✓	Ultra-lightweight early physics CNN.
DehazeFormer-T [5]	TIP '23	2.73	✓	Highly efficient Transformer baseline.
DEA-Net [6]	TIP '24	16.44	✓	Recent SOTA data-driven CNN architecture.
FFA-Net [4]	AAAI '20	24.68	✓	Heavy feature-fusion attention anchor.
Excluded Models: Recent Heavyweight Architectures (2025–2026)				
UVM-Net	CVPRW '25	173.55	✗	Massive computational cost (3× to 8× higher than our heaviest evaluated baseline); fundamentally incompatible with strict edge NPU power budgets.
DFFNet	Remote Sens. '25	168.22	✗	
Ksformer	arXiv '25	92.12	✗	
Excluded Models: Generative / Diffusion Approaches				
HazeDiff	CVPR '24	~500.00 [†]	✗	Iterative multi-step sampling inherently violates real-time latency constraints (> 30 FPS). Requires desktop-class VRAM (e.g., 24 GB) rather than edge SoCs.
DehazeDDPM	CVPR '25		✗	
IR-SDE / FrDiff	ICML '22 / CVPR '25	Not Reported	✗	

† Effective compute estimated assuming ≥ 50 denoising steps through a standard U-Net backbone at 256×256 .

D Architecture Details

Each encoder stage consists of two DWSep blocks with a residual shortcut, followed by $2\times$ max-pooling; the four stages produce feature maps at H , $H/2$, $H/4$, and $H/8$ (bottleneck). The image branch accepts a 3-channel input (monochrome NIR replicated or colour); the depth branch accepts 2

channels (sparse depth D_s and binary validity mask M). A Squeeze-and-Excitation (SE) channel attention block [14] with reduction ratio 4 is applied once at the bottleneck after gated fusion, recalibrating channel responses before decoding. The decoder consists of three upsampling stages, each performing bilinear $2\times$ upsampling followed by channel-wise concatenation with the corresponding fused skip features and refinement through a DWSep residual block. The bottleneck fused features ($8c$ channels at $H/8$) serve as the decoder input; skip connections from scales 3, 2, and 1 progressively reintroduce spatial detail. The head projections use 1×1 convolutions throughout; airlight is predicted via global average pooling of the bottleneck followed by a 3-unit fully-connected layer.

E Training Details and Hyperparameters

All PhysDNet variants share a common training recipe and loss weights, differing only in base channel width c . The **L** variant additionally enables mixed-precision (AMP) to fit within VRAM at $c=96$.

Table 8: **Training recipe.** Shared across all variants except where noted. [†] AMP enabled for PhysDNet-L only.

Hyperparameter	PhysDNet-S	PhysDNet-M	PhysDNet-L
Base channel width c	32	64	96
SE attention	✓	✓	✓
Epochs	500	500	500
Batch size	16	16	8 [†]
Optimizer	AdamW		
Learning rate	2×10^{-4}		
Weight decay	1×10^{-4}		
LR scheduler	Cosine annealing $\rightarrow 10^{-6}$		
Warmup epochs	10		
Gradient clipping	$\ \nabla\ _2 \leq 1.0$		
Crop size	256×256		
Data augmentation	Random flip + crop		
Mixed precision (AMP)	×	×	✓
β range (haze synthesis)	$\mathcal{U}(0.005, 0.04)$		
A_k range (airlight per channel)	$\mathcal{U}(0.7, 1.0)$		
Max LiDAR depth	120 m		

Table 9: **Loss component weights** (λ). All variants share the same loss weights and contrastive activation schedule (epoch 185); they differ only in base channel width c .

Data-driven / Physics / Auxiliary shading indicates loss family.

Loss	Symbol	Supervises	S	M	L	Role
Reconstruction	$\mathcal{L}_{\text{rec}}$	$\hat{J}_{\text{dir}}$ vs. J_{gt}	1.0	1.0	1.0	Primary restoration
Perceptual	$\mathcal{L}_{\text{perc}}$	VGG-16 features	0.05	0.05	0.05	Feature-space similarity
Gradient	$\mathcal{L}_{\text{grad}}$	$\nabla \hat{J}_{\text{dir}}$ vs. ∇J_{gt}	0.5	0.5	0.5	Edge preservation
Physics recon.	$\mathcal{L}_{\text{pr}}$	$\hat{J}_{\text{phys}}$ vs. J_{gt}	1.0	1.0	1.0	Physics-head fidelity
Transmission	$\mathcal{L}_t$	$\hat{t}$ vs. t_{gt}	2.0	2.0	2.0	LiDAR-supervised t
Smoothness	$\mathcal{L}_{\text{sm}}$	$\nabla \hat{t}$	0.1	0.1	0.1	Edge-aware regularization
Consistency	$\mathcal{L}_{\text{con}}$	$\hat{J}_{\text{dir}}$ vs. $\hat{J}_{\text{phys}}$	0.2	0.2	0.2	Dual-head agreement
FFT	$\mathcal{L}_{\text{fft}}$	Frequency domain	0.1	0.1	0.1	High-frequency detail
Contrastive	$\mathcal{L}_{\text{contra}}$	Latent features	0.1	0.1	0.1	Physics-aware bound
start epoch			185	185	185	Delayed activation

Table 10: **PhysDNet model family.** Scaling the base channel width c yields three variants spanning the accuracy–efficiency Pareto front. All variants share the same dual-encoder, gated-fusion, SE-decoder topology (§2.1) and dual-head design (§2.2). Physics head parameters are excluded from the inference count as only the direct head is deployed.

Variant	c	SE Attn.	Params (M)	GMACs	Bottleneck	Target
PhysDNet-S	32	✓	0.72	2.24	$256 \times \frac{H}{8} \times \frac{W}{8}$	Ultra-low-power NPU
PhysDNet-M	64	✓	2.95	8.56	$512 \times \frac{H}{8} \times \frac{W}{8}$	Edge GPU / mid-range NPU
PhysDNet-L	96	✓	6.59	18.98	$768 \times \frac{H}{8} \times \frac{W}{8}$	High-perf. edge accelerator

Convergence and Training Budgets. All models were trained until performance plateaued under their respective training configurations, subject to a fixed computational budget. In practice, we observe that PSNR improvements saturate after approximately 80 epochs for most baselines, with additional training yielding only marginal gains (typically < 0.3 dB). This behavior is consistent across all evaluated methods, indicating that extended training primarily increases compute cost without significantly affecting relative performance rankings. PhysDNet variants were trained for 500 epochs to ensure the complex multi-term objective (including late-activated contrastive losses) fully converged, though primary restoration metrics similarly plateaued early in the schedule. Experiments require 24 GB VRAM GPU.

F Gradient Weighting under the Scattering Model: Full Derivation

Section 2.3 empirically characterised the $(\hat{t} + \epsilon)^{-2}$ gradient structure induced by differentiating the Koschmieder inversion and briefly noted its structural correspondence to the inversion’s noise variance. This appendix provides the formal derivations of this variance-gradient relationship, alongside the airlight-gradient analysis, the bounded-gradient discussion, and a scope statement delineating what the analysis does and does not claim.

Setup. We adopt the standard atmospheric scattering model with additive sensor noise [20, 31, 32]:

$$I(x) = J(x)t(x) + A(1 - t(x)) + \xi(x), \quad \xi(x) \sim \mathcal{N}(0, \sigma^2), \quad (8)$$

with $\xi(x)$ independent across pixels. Given estimates $\hat{t}(x), \hat{A}$, the physics head computes the ϵ -stabilised inversion

$$\hat{J}_{\text{phys}}(x) = \frac{I(x) - \hat{A}(1 - \hat{t}(x))}{\hat{t}(x) + \epsilon}, \quad (9)$$

consistent with Eq. (5) in the main text.

Proposition 1 (Transmission-Scaled Noise Variance). *Under Eq. (8) and first-order estimates $\hat{t}(x) = t(x) + \delta_t$, $\hat{A} = A + \delta_A$ with $|\delta_t|, |\delta_A| = O(\delta)$ and $\delta \ll t_{\min}$, the inverted radiance admits*

$$\hat{J}_{\text{phys}}(x) = J(x) + \frac{\xi(x)}{t(x)} + O(\delta), \quad \text{Var}[\hat{J}_{\text{phys}}(x)] = \frac{\sigma^2}{t(x)^2} + O(\delta). \quad (10)$$

Proof. Substituting Eq. (8) into Eq. (9):

$$\hat{J}_{\text{phys}} = \frac{Jt + A(1 - t) + \xi - (A + \delta_A)(1 - (t + \delta_t))}{(t + \delta_t) + \epsilon}.$$

Expanding the bracket in the numerator gives $(A + \delta_A)(1 - t - \delta_t) = A(1 - t) - A\delta_t + \delta_A(1 - t) - \delta_A\delta_t$, so the $A(1 - t)$ terms cancel and

$$\text{numerator} = Jt + \xi + A\delta_t - \delta_A(1 - t) + O(\delta^2).$$

For the denominator, since $\delta, \epsilon \ll t$, we have the first-order expansion

$$\frac{1}{t + \delta_t + \epsilon} = \frac{1}{t} \cdot \frac{1}{1 + (\delta_t + \epsilon)/t} = \frac{1}{t} - \frac{\delta_t + \epsilon}{t^2} + O(\delta^2, \epsilon^2).$$

Multiplying numerator and denominator and retaining terms up to $O(\delta)$:

$$\begin{aligned}\hat{J}_{\text{phys}} &= [Jt + \xi + A\delta_t - \delta_A(1-t)] \cdot \frac{1}{t} - Jt \cdot \frac{\delta_t + \epsilon}{t^2} + O(\delta^2, \epsilon) \\ &= J + \frac{\xi}{t} + \frac{A\delta_t - \delta_A(1-t)}{t} - \frac{J\delta_t}{t} + O(\delta^2, \epsilon) \\ &= J + \frac{\xi}{t} - \frac{(J-A)\delta_t + (1-t)\delta_A}{t} + O(\delta^2, \epsilon),\end{aligned}$$

where the cross-terms (e.g. $\xi \cdot O(\delta, \epsilon)$, $A\delta_t \cdot O(\delta, \epsilon)$) all collapse into $O(\delta^2, \epsilon)$. The deterministic perturbation terms are $O(\delta)$; only ξ/t contributes stochastic variance. Since $\xi \sim \mathcal{N}(0, \sigma^2)$, $\text{Var}[\hat{J}_{\text{phys}}] = \sigma^2/t^2 + O(\delta)$. $\square$

Proposition 2 (Structured Jacobian Scaling). The Jacobian of the inversion with respect to transmission is

$$\frac{\partial \hat{J}_{\text{phys}}}{\partial \hat{t}}(x) = \frac{\hat{A} - I(x)}{(\hat{t}(x) + \epsilon)^2}, \quad (11)$$

treating $(I, \hat{A})$ as fixed with respect to $\hat{t}$.

Derivation. Write $\hat{J}_{\text{phys}} = u/v$ with $u(\hat{t}) = I - \hat{A}(1 - \hat{t}) = I - \hat{A} + \hat{A}\hat{t}$ and $v(\hat{t}) = \hat{t} + \epsilon$. Then $\partial u/\partial \hat{t} = \hat{A}$, $\partial v/\partial \hat{t} = 1$, and the quotient rule yields

$$\frac{\partial \hat{J}_{\text{phys}}}{\partial \hat{t}} = \frac{\hat{A}(\hat{t} + \epsilon) - [I - \hat{A}(1 - \hat{t})]}{(\hat{t} + \epsilon)^2} = \frac{\hat{A} - I + \hat{A}\epsilon}{(\hat{t} + \epsilon)^2} = \frac{\hat{A} - I}{(\hat{t} + \epsilon)^2} + O(\epsilon),$$

which gives Eq. (11) after dropping the $O(\epsilon)$ term consistent with the main text. $\square$

This expression reveals a *structural* inverse-square dependence on transmission through the denominator $(\hat{t} + \epsilon)^{-2}$. Consequently, the computational graph intrinsically amplifies gradients in low-transmission (dense fog) regions. The effective gradient magnitude additionally depends on the residual term $(\hat{A} - I(x))$, which introduces a data-dependent modulation. Under the scattering model, $(\hat{A} - I(x)) \approx t(x)(A - J(x))$ when estimates are accurate, partially attenuating the scaling in well-reconstructed regions. Nevertheless, the dominant $(\hat{t} + \epsilon)^{-2}$ structure ensures that low-transmission pixels receive substantially larger gradient magnitudes throughout training. Importantly, this amplification arises directly from differentiating the physical inversion and does not rely on explicit weighting heuristics, auxiliary parameters, or curriculum schedules.

Corollary (Loss-gradient scaling). Under $\mathcal{L}_{\text{pr}} = \|\hat{J}_{\text{phys}} - J_{\text{gt}}\|_1$,

$$\frac{\partial \mathcal{L}_{\text{pr}}}{\partial \hat{t}(x)} = \text{sign}(\hat{J}_{\text{phys}}(x) - J_{\text{gt}}(x)) \cdot \frac{\hat{A} - I(x)}{(\hat{t}(x) + \epsilon)^2}, \quad (12)$$

and under an ℓ_2 loss the sign is replaced by $2(\hat{J}_{\text{phys}} - J_{\text{gt}})$. In both cases the amplification propagates unchanged into the encoder weights via $\partial \hat{t}/\partial \theta_{\text{enc}}$.

Airlight gradient: physically correct supervision routing. The magnitude of the airlight gradient is $|\partial \hat{J}_{\text{phys}}/\partial \hat{A}| = |1 - \hat{t}|/(\hat{t} + \epsilon)$, which scales as $\hat{t}^{-1}$ for small $\hat{t}$ and vanishes as $\hat{t} \rightarrow 1$. The network therefore learns to estimate airlight primarily from foggy regions, where atmospheric light dominates the observation, and ignores clear-air pixels that carry no information about A . This supervision routing matches the physical reality: airlight is only observable where scattering is present.

Empirical spatial validation. Figure 6 confirms the $(\hat{t} + \epsilon)^{-2}$ scaling holds on real (non-synthetic) STF fog frames at the per-pixel level: the analytically predicted gradient magnitude map spatially aligns with low-transmission regions in the predicted $\hat{t}$, with high-gradient activations concentrated precisely where the physics model is most sensitive.

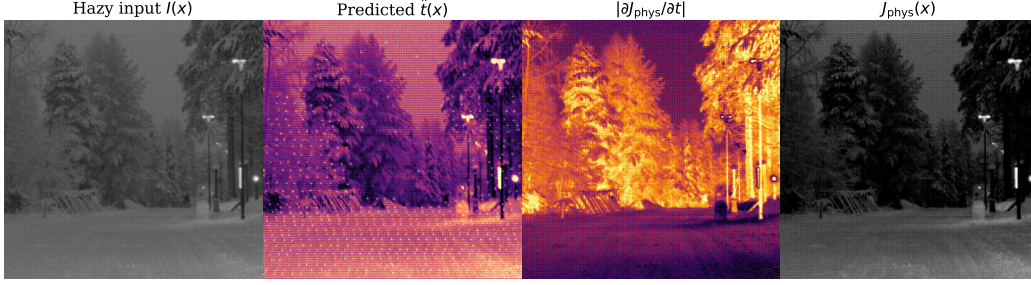


Figure 6: Empirical gradient map on a real STF fog frame. Left to right: hazy input, predicted transmission $\hat{t}$, analytical gradient magnitude map $|\partial \hat{J}_{\text{phys}} / \partial \hat{t}|$, and physics-restored output. High-gradient regions align spatially with low-transmission regions, empirically validating the density-adaptive behaviour predicted by Propositions 1–2.

Contrast with the direct-head gradient. The direct head’s gradient through the shared encoder, $\partial \hat{J}_{\text{dir}} / \partial \mathbf{z} = \sigma'(\cdot) W_{\text{dir}}$, is spatially uniform and has no notion of local fog density. The physics head provides spatially structured, density-adaptive magnitudes. Both signals flow into the same encoder, but the physics path shapes features toward physically meaningful representations even though only the direct head is used at inference.

Interpretive connection to inverse-variance weighting. Proposition 1 establishes that the stochastic sensitivity of the inversion scales as t^{-2} . Classical heteroscedastic regression [19, 33, 34] exploits this inversely, multiplying each sample’s loss by $1 / \text{Var}[\hat{J}] \propto t^2$ to down-weight unreliable pixels. Differentiating through Eq. (9) applies the structural inverse, up-weighting low-transmission regions. The two schemes are not in conflict; they operate on different signal models. Inverse-variance weighting is optimal when the signal is dominated by irreducible noise. Under the scattering model, the numerator $(\hat{A} - I(x)) = -(J - A)t + O(\xi)$ is dominated by the deterministic $(J - A)$ term in degraded regions where $(J - A)$ is large, so larger updates reduce estimation error rather than fit noise. We interpret the physics head as performing implicit hard-example mining along the transmission axis, empirically supported by the NoPhy ablation (§3.3).

Bounded gradients and stability. To ensure numerical stability, the transmission is constrained to $\hat{t} \in [t_{\min}, t_{\max}]$ with $t_{\min} = 0.05$ (Eq. 3). This bounds the Jacobian:

$$\left| \frac{\partial \hat{J}_{\text{phys}}(x)}{\partial \hat{t}} \right| \leq \frac{|\hat{A} - I(x)|}{(t_{\min} + \epsilon)^2} \leq \frac{1}{(t_{\min} + \epsilon)^2}, \quad (13)$$

since $\hat{A}, I(x) \in [0, 1]$. In practice, this upper bound is approximately 400, and is further controlled via gradient clipping (Appendix E). As a result, the optimization remains within a bounded-gradient regime compatible with standard convergence guarantees for stochastic gradient methods [35, 36].

Implicit curriculum. The transmission-dependent gradient scaling induces an implicit curriculum [37] over training samples. Early in training, low-transmission (high-degradation) regions dominate the gradient signal due to their larger magnitude. As the model improves and transmission estimates stabilize, the gradient distribution naturally rebalances without requiring explicit scheduling. This behavior emerges directly from the physics-based formulation and does not introduce additional hyperparameters.

Scope. The analysis assumes (i) homogeneous single-scattering, (ii) Gaussian i.i.d. sensor noise, and (iii) first-order accuracy of $(\hat{t}, \hat{A})$. Real fog may violate (i) through non-uniform airlight and multiple scattering; low-light imagery may violate (ii) through Poisson-dominated shot noise; heavily degraded regions may violate (iii) when $\delta \not\ll t_{\min}$. The structural amplification arises directly from the $1/\hat{t}$ factor in Eq. (9), not from Gaussianity—and is therefore robust under (ii); the prefactor may shift under (i) and (iii). The cross-domain results on O-HAZE and I-HAZE (§3.6), where

Table 11: **PhysDNet-M INT8 quantization quality** on the fixed-function NPU. PSNR drop vs. FP32 is ~ 1.1 dB within the training β range. $\pm$ values are 1σ over n samples.

β	n	INT8 PSNR (dB)	FP32 PSNR (dB)
0.005	29	35.30 $\pm$ 4.9	37.95 $\pm$ 4.9
0.010	29	36.38 $\pm$ 3.9	37.19 $\pm$ 5.1
0.020	29	34.08 $\pm$ 5.8	35.58 $\pm$ 5.9
0.040	29	31.42 $\pm$ 6.1	32.34 $\pm$ 6.5
<i>In-distribution avg</i>		34.30	35.77
0.060	28	22.78 $\pm$ 7.9	25.91 $\pm$ 8.4
0.080	28	20.07 $\pm$ 6.1	20.02 $\pm$ 6.4
0.100	28	14.31 $\pm$ 5.2	16.18 $\pm$ 6.0
<i>Overall avg</i>		27.89	29.31

the NoPhy ablation degrades PSNR-Y by 1.53 and 0.60 dB on real non-synthetic haze, provides empirical evidence that the mechanism transfers beyond the assumed regime.

G Edge Deployment: Full Results

This appendix provides the full deployment measurements summarized in §3.8: INT8 quantization quality on the fixed-function NPU, qualitative comparison under quantization, and the full embedded-GPU benchmark including CUDA FP32 reference, numbers and onboard power profiles.

G.1 INT8 Quantization Quality on the Fixed-Function NPU

To verify that INT8 post-training quantization preserves restoration quality, we study PhysDNet-M in detail. We use the vendor compiler’s default calibration pipeline with 64 real STF samples ($\beta=0.02$, $A=0.8$). **We do not apply quantization-aware training, mixed-precision layer policies, custom activation-range search, or any other vendor-specific optimization;** the reported gap is therefore representative of naive post-training quantization rather than a tuned result. Evaluation covers 200 held-out samples spanning $\beta \in \{0.005, \dots, 0.10\}$ (Table 11).

Within the training range ($\beta \leq 0.04$), INT8 incurs only a ~ 1.1 dB average PSNR drop vs. FP32; at $\beta=0.01$ the quantized model comes within 0.8 dB of FP32. Out-of-distribution degradation ($\beta > 0.04$) follows the same graceful decline as FP32, confirming that the physics-guided head is not destabilized by aggressive quantization.

G.2 Embedded-GPU Full Benchmark

Table 12 reports the full benchmark under both CUDA FP32 and TensorRT FP16 execution providers on the Jetson Orin Nano. The TensorRT FP16 path yields $3\text{--}5\times$ speedups over CUDA FP32 across the board with no measurable PSNR degradation, and is therefore the reference deployment configuration in the main text.

Hardware disclosure. The NVIDIA Jetson Orin Nano Super (8 GB RAM, MAXN power mode) is named explicitly. The fixed-function NPU target is anonymized in the submission version for double-blind review; full vendor and model identification will be provided in the camera-ready version. Power and throughput figures are otherwise reported as measured.

H Baseline Descriptions

We compare PhysDNet against both established dehazing architectures and architectural ablations to quantify the contribution of each design choice.

Published baselines. **AOD-Net** [3]: a lightweight all-in-one CNN that directly regresses the clean image, representing early physics-reformulated dehazing. **FFA-Net** [4]: a feature-fusion attention

Table 12: **Full embedded-GPU benchmark** (256×256 , 50 warmup + 300 timed iterations, MAXN mode). Mean and median latency, FPS, and onboard rail power reported for both CUDA (FP32) and TensorRT FP16 providers.

Model	Params	Provider	Mean (ms)	Median (ms)	FPS	$\bar{P}$ (W)	$P_{\max}$ (W)
AOD-Net LiDAR	0.002 M	TRT_FP16	3.06	3.05	327.0	4.94	5.21
AOD-Net LiDAR	0.002 M	CUDA	12.36	12.35	80.9	5.13	5.66
FFA-Net LiDAR	0.77 M	TRT_FP16	21.31	21.30	46.9	8.21	10.09
FFA-Net LiDAR	0.77 M	CUDA	83.67	83.65	12.0	13.39	15.44
DehazeFormer-T LiDAR	1.36 M	TRT_FP16	21.25	21.24	47.1	12.38	12.93
DehazeFormer-T LiDAR	1.36 M	CUDA	62.61	62.60	16.0	11.47	13.17
DEA-Net LiDAR	3.65 M	TRT_FP16	35.12	35.12	28.5	10.17	12.42
DEA-Net LiDAR	3.65 M	CUDA	89.93	89.93	11.1	12.45	13.05
PhysDNet-S (ours)	0.72 M	TRT_FP16	9.24	9.23	108.3	13.45	13.48
PhysDNet-S (ours)	0.72 M	CUDA	40.72	40.70	24.6	10.63	13.54
PhysDNet-M (ours)	2.95 M	TRT_FP16	17.09	17.09	58.5	12.38	12.42
PhysDNet-M (ours)	2.95 M	CUDA	79.08	79.07	12.6	12.78	14.97
PhysDNet-L (ours)	6.59 M	TRT_FP16	25.80	25.80	38.8	11.80	13.01
PhysDNet-L (ours)	6.59 M	CUDA	120.52	120.54	8.3	13.33	16.12

Table 13: **Full efficiency breakdown**. Latency and FPS on a single NVIDIA GPU, batch size 1. [†]Measured at 256×256 . We report the PyTorch numbers (no additional optimization is conducted)

Model	Config	Params (M)	GMACs	GFLOPs	Lat. (ms)	FPS
AOD-Net	NIR	0.002	0.23	0.46	0.3	3 753
AOD-Net	LiDAR	0.002	0.24	0.47	0.3	3 550
DehazeFormer-T	NIR	1.36	10.92	21.83	7.0	143
DehazeFormer-T	LiDAR	1.36	10.97	21.95	7.0	143
FFA-Net	NIR	0.77	98.57	197.15	11.5	87
FFA-Net	LiDAR	0.77	98.73	197.45	11.5	87
DEA-Net	RGB	3.65	16.42	32.84	8.9	112.6
DEA-Net	LiDAR	3.65	16.44	32.88	8.9	112.6
PhysDNet-S [†]	ch32	0.72	2.24	4.47	1.5	661
PhysDNet-M [†]	ch64, SE	2.95	8.56	17.13	2.3	429
PhysDNet-L [†]	ch96, SE	6.59	18.98	37.97	3.6	275

CNN; strong attention-based baseline. **DehazeFormer-T** [5]: a Transformer-based restoration model representing modern attention architectures. **DEA-Net** [6]: a difference-aware encoder-decoder network. We include it to benchmark our physics-guided feature extraction against, purely data-driven detail enhancement.

I Model Efficiency Breakdown

Table 13 provides a detailed efficiency comparison across all models. PhysDNet variants are measured at 256×256 ; baselines at 512×512 (their best-performing resolution per Table 1).

J Auxiliary Loss Ablation

We ablate the two auxiliary loss components, FFT and Contrastive, that supplement the base objective (reconstruction + perceptual + gradient). We trained PhysDNet-M (ch64) for 200 epochs at 512×512 ; each row in Table 14 enables one auxiliary loss on top of the base objective. The results motivate the inclusion of both in the complete recipe (table 9), which combines them under the extended 500-epoch schedule used in the main results.

Table 14: **Auxiliary loss ablation.** PhysDNet-M (ch64) trained for 200 epochs at 512×512 . Each row enables one auxiliary loss on top of the base (reconstruction + perceptual + gradient) objective. FFT contributes the largest PSNR gain; Contrastive contributes the largest SSIM gain.

Auxiliary loss	PSNR (dB) $\uparrow$	SSIM $\uparrow$
Base objective only	32.38	0.9315
+ Contrastive	32.97	0.9406
+ FFT	33.02	0.9401

K Downstream Detection Protocol

Fog generation. Synthetic fog is applied using the same Koschmieder model as in training (§3.1), with $\beta=0.06$ and $A=0.8$ to test at out-of-distribution severity.

Detector and classes. YOLOv8-M [21] (COCO-pretrained, $\text{conf} \geq 0.25$, $\text{IoU} \geq 0.5$). STF KITTI-format labels are mapped to COCO classes and grouped into three families: *vehicle* (car, bus, truck), *person*, and *cyclist*. AP per family uses 101-point interpolation (COCO style).

Tiled inference. Dehazing at 256×256 with overlapping tiles (stride 192), linearly blended in the overlap to avoid boundary artifacts.

L Cross-Domain Generalization: Additional Details

Chrominance recovery. The restored output on O-HAZE is structurally strong but desaturated. We use the model’s predicted transmission $\hat{t}$ for *transmission-guided chrominance transfer*: convert both restored and hazy images to YCbCr, keep the Y channel from the restored output, and transplant attenuated Cb/Cr from the hazy input:

$$\text{Cb}_{\text{out}} = 0.5 + \hat{t} \cdot (\text{Cb}_{\text{hazy}} - 0.5), \quad \text{Cr}_{\text{out}} = 0.5 + \hat{t} \cdot (\text{Cr}_{\text{hazy}} - 0.5).$$

This attenuates haze-coloured chrominance in heavily fogged regions (low $\hat{t}$) while preserving natural colour where the scene is clear (high $\hat{t}$).

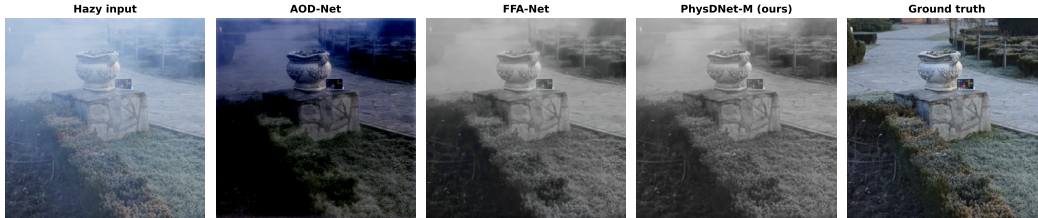


Figure 7: **Zero-shot O-HAZE restoration.** Left to right: hazy input, AOD-Net, FFA-Net, PhysDNet-M (ours), ground truth. All models trained on STF only.

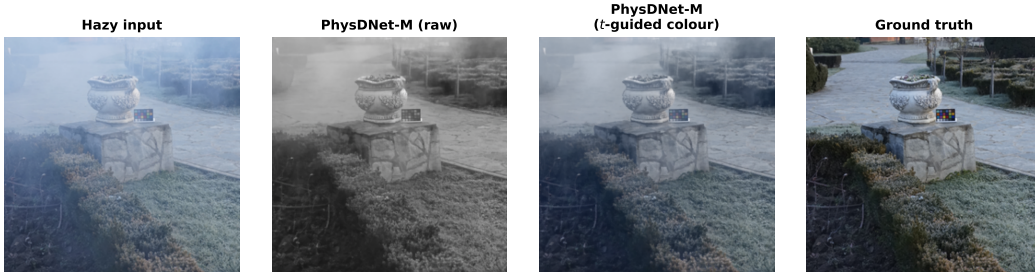


Figure 8: **Transmission-guided chrominance recovery.** Left to right: hazy input, raw PhysDNet-M output, after $\hat{t}$ -guided chrominance transfer, ground truth.

M Cross-Dataset Generalization: NH-HAZE

NH-HAZE [38] contains 55 *non-homogeneous* real outdoor hazy/clean image pairs (NTIRE 2020). Unlike O-HAZE’s uniform fog, NH-HAZE exhibits spatially varying haze density, directly testing whether the physics head transfers despite the Koschmieder model’s homogeneous-scattering assumption. As with other cross-dataset benchmarks, we feed $\mathbf{d}=\mathbf{0}$, $\mathbf{m}=\mathbf{0}$ and resize to 256×256 . No fine-tuning.

Table 15: **Zero-shot NH-HAZE results at 256×256** . All models trained on STF; no NH-HAZE fine-tuning. Best in **bold**.

Method	PSNR-Y $\uparrow$	SSIM-Y $\uparrow$	PSNR-RGB $\uparrow$	SSIM-RGB $\uparrow$
AOD-Net	12.21	0.3835	12.12	0.3791
FFA-Net	14.54	0.5037	14.16	0.4889
DehazeFormer-T	14.55	0.4889	14.18	0.4742
DEA-Net	14.59	0.5069	14.24	0.4932
PhysDNet-M (NoPhy)	14.20	0.5294	13.98	0.5165
PhysDNet-S (ours)	14.70	0.4950	14.37	0.4807
PhysDNet-M (ours)	14.74	0.5111	14.37	0.4965
PhysDNet-L (ours)	14.45	0.5159	14.09	0.5012

PhysDNet-M achieves the highest PSNR-Y (14.74 dB), outperforming DEA-Net by +0.15 dB and FFA-Net by +0.20 dB; PhysDNet-L attains the best SSIM-Y (0.5159). PhysDNet-S remains competitive (14.70 dB) at $4\times$ fewer parameters than the M variant. Removing the physics head (NoPhy) reduces PSNR-Y by 0.54 dB and drops the model below all modern baselines, confirming that the physics-head contribution transfers to non-homogeneous fog despite the underlying Koschmieder model assuming homogeneous scattering. This evidence directly addresses the model-mismatch concern raised in our Limitations.

N LiDAR Sparsity Robustness

Automotive LiDAR produces inherently sparse depth maps—typically $<5\%$ of pixels carry valid returns, with sparsity varying significantly across weather conditions, surface reflectivity, and sensor occlusion. To study how PhysDNet’s restoration quality degrades as depth information becomes scarcer, we systematically subsample valid LiDAR returns from 100% down to 0% (pure NIR fallback) while keeping all other inputs fixed. This stress-tests the gated-fusion mechanism (§2.1): a well-designed gate should learn to fall back to NIR features when depth is absent ($\mathbf{m}=\mathbf{0}$), rather than catastrophically failing.

Protocol. For each keep ratio $r \in \{1.0, 0.75, 0.50, 0.25, 0.10, 0.05, 0.01, 0.0\}$, we randomly retain r fraction of valid mask pixels in the STF validation set (1 299 images) and zero-out the corresponding depth entries. The model receives a hazy image $I \in [0, 1]^{3\times H\times W}$, sparse LiDAR depth D_s , and validity mask M , and jointly predicts a restored image, transmission map, and airlight vector.

Table 16: **LiDAR sparsity ablation** on STF val (256×256). PSNR (dB) at each keep ratio. At $r=0$ (no LiDAR) the model falls back to NIR-only restoration via the learned gated-fusion mechanism. Training uses $r\approx 3\%$ (typical Velodyne HDL-64 density).

Model	100%	75%	50%	25%	10%	5%	1%	0%
PhysDNet-S	32.80	31.99	30.98	29.77	28.60	28.03	27.29	26.52
PhysDNet-M	35.51	34.44	33.27	32.17	31.16	30.66	30.23	29.70
PhysDNet-L	36.54	34.64	32.50	31.15	30.73	30.55	30.69	30.57
PhysDNet-M (NoPhy)	30.02	29.38	28.73	28.06	27.41	27.00	26.06	25.60

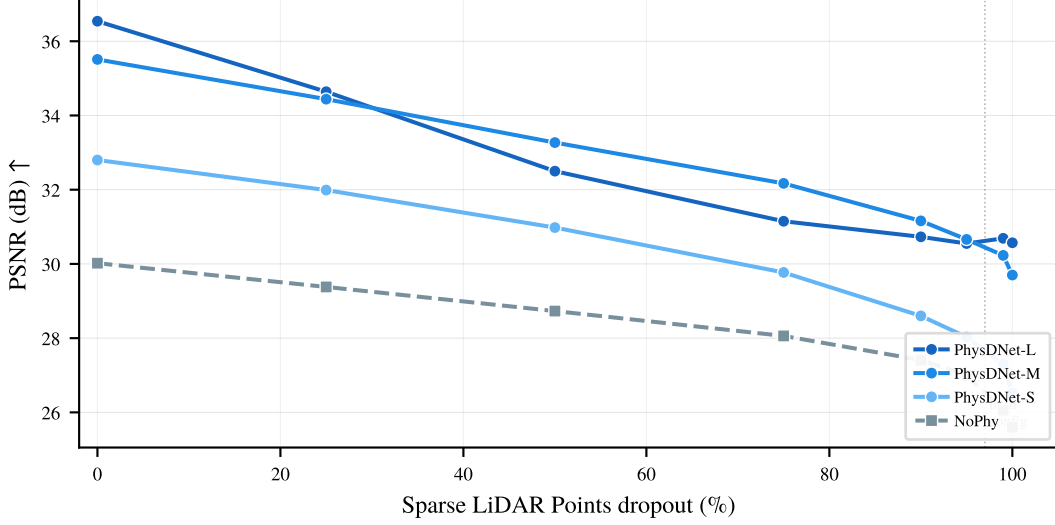


Figure 9: **PSNR vs. LiDAR keep ratio.** All PhysDNet variants degrade gracefully as LiDAR points are removed. Even at 0% density (NIR only), PhysDNet-M retains a +4.10 dB advantage over the NoPhy ablation, confirming that the physics-guided representation transfers across input modalities. Vertical dotted line marks the typical training-time density ($\sim 3\%$).

Three observations. (i) All PhysDNet variants degrade smoothly rather than collapsing as LiDAR density decreases, confirming the gated-fusion mechanism (§2.1) successfully learned to attribute weight to NIR features when depth signal is unavailable. (ii) Even at 0% LiDAR (pure NIR fallback), PhysDNet-M retains 29.70 dB—still 4.10 dB above the NoPhy ablation at matched 0% density (25.60 dB). This confirms that the physics head’s contribution is not merely depth-driven but reshapes the encoder’s NIR processing as well. (iii) At extreme sparsity ($r=1\%$, well below training density), PhysDNet-L’s PSNR (30.69) remains higher than at $r=0\%$ (30.57), indicating that even one valid LiDAR point per 100 pixels provides usable signal. This degradation profile validates deployment under realistic LiDAR conditions where weather can drop point density significantly below nominal.

O No-Reference Perceptual Quality

Models trained on STF synthetic fog produce near-achromatic restorations on cross-dataset images (§3.6), so chrominance-sensitive full-reference metrics underestimate restoration quality. We supplement Table 4 with two established no-reference IQA metrics: **NIQE** [39] and **BRISQUE** [40] (lower = better; both assess naturalness and distortion blindly without ground truth). We evaluate on O-HAZE (45 outdoor pairs, no LiDAR; $d=0$).

Table 17: **No-reference perceptual quality on O-HAZE** (NIQE / BRISQUE, lower = better). Numbers in parentheses are 1σ across the 45-image O-HAZE validation set. “Hazy” is the unprocessed input. Best in **bold**.

Method	NIQE ↓		BRISQUE ↓	
Hazy input	4.33	($\sigma=1.14$)	7.75	($\sigma=7.02$)
AOD-Net	6.37	($\sigma=4.18$)	20.62	($\sigma=15.58$)
FFA-Net	4.30	($\sigma=1.41$)	12.25	($\sigma=8.84$)
DehazeFormer-T	4.32	($\sigma=1.25$)	12.76	($\sigma=9.83$)
DEA-Net	4.64	($\sigma=1.62$)	11.95	($\sigma=10.66$)
PhysDNet-S	4.21	($\sigma=1.04$)	15.98	($\sigma=6.92$)
PhysDNet-M	3.92	($\sigma=1.08$)	11.13	($\sigma=5.58$)
PhysDNet-L	4.06	($\sigma=1.10$)	9.49	($\sigma=6.73$)
PhysDNet-M (NoPhy)	3.91	($\sigma=1.27$)	11.96	($\sigma=6.29$)

Table 18: **STF dataset split statistics.** 10 280 frames with weather metadata, Feb 2018–May 2019, -25.5°C to 19.7°C .

Condition	Frames	Ratio	Usage
Clear	1 497	14.6%	Synthetic haze training
Overcast	2 715	26.4%	Synthetic haze training
Light Fog	3 788	36.8%	Real-fog eval (Table 5)
Dense Fog	1 219	11.9%	Real-fog eval (Table 5)
Snow/Ice	1 061	10.3%	Reserved (future work)
Total	10 280	100%	

Discussion. PhysDNet leads or matches the strongest baselines on both no-reference metrics. PhysDNet-M achieves the best NIQE among learned methods (3.92, vs. DEA-Net 4.64 and AOD-Net 6.37); PhysDNet-L achieves the best BRISQUE (9.49, vs. FFA-Net 12.25 and DEA-Net 11.95). Compared to the NoPhy ablation, PhysDNet-M shows a consistent BRISQUE improvement (-0.83), while NIQE differences fall within noise—NIQE is a weaker discriminator than BRISQUE for low-distortion images, where most methods already approach naturalness statistics. Together with the PSNR-Y and SSIM-Y results in Table 4, these no-reference metrics provide independent confirmation that PhysDNet’s restoration quality on real haze is not an artifact of synthetic-data circularity.

P Dataset Details

The Seeing Through Fog (STF) benchmark [1] provides 10 280 synchronised multi-sensor frames captured over 14 months of real adverse weather across Northern and Central Europe, spanning a temperature range of -25.5°C to 19.7°C . Each frame includes monochrome 12-bit gated NIR imagery, sparse Velodyne HDL-64 LiDAR returns, RGB stereo, and weather metadata. We use only the gated NIR and LiDAR modalities, as they are the most informative under fog (§B). Table 18 summarises the per-condition split and how each subset is used in our experiments. Figure 10 illustrates the multi-modal perception challenge: LiDAR returns are sparse and non-uniform across weather conditions, motivating our gated-fusion design (§2.1).

Q Additional Qualitative Results

We provide additional dual-head visualisations to illustrate the complementarity discussed in §2.2. Figure 11 shows side-by-side outputs of the direct head and the physics head across eight diverse synthetic-fog samples from the STF validation set, alongside the predicted transmission map and sparse LiDAR overlay. The physics head produces sharper outputs in well-conditioned regions but exhibits noise amplification under dense haze (blue boxes in fig. 11), where the $(\hat{t} + \epsilon)^{-2}$ inversion is most ill-conditioned. The unconstrained direct head, trained jointly under the physics-anchored objective (§2.4), successfully suppresses these artifacts while inheriting the density-adaptive representation shaped during training. Only the direct head is deployed at inference.

R Out-of-Distribution Fog Robustness

To evaluate model robustness to out-of-distribution weather severities, we sweep the scattering coefficient $\beta \in [0.005, 0.10]$ at fixed global airlight ($A = 0.8$) across the full 1,299-image validation set. As shown in fig. 12, values $\beta \geq 0.05$ fall entirely outside the training distribution.

STF Dataset — Camera + LiDAR per Weather Condition

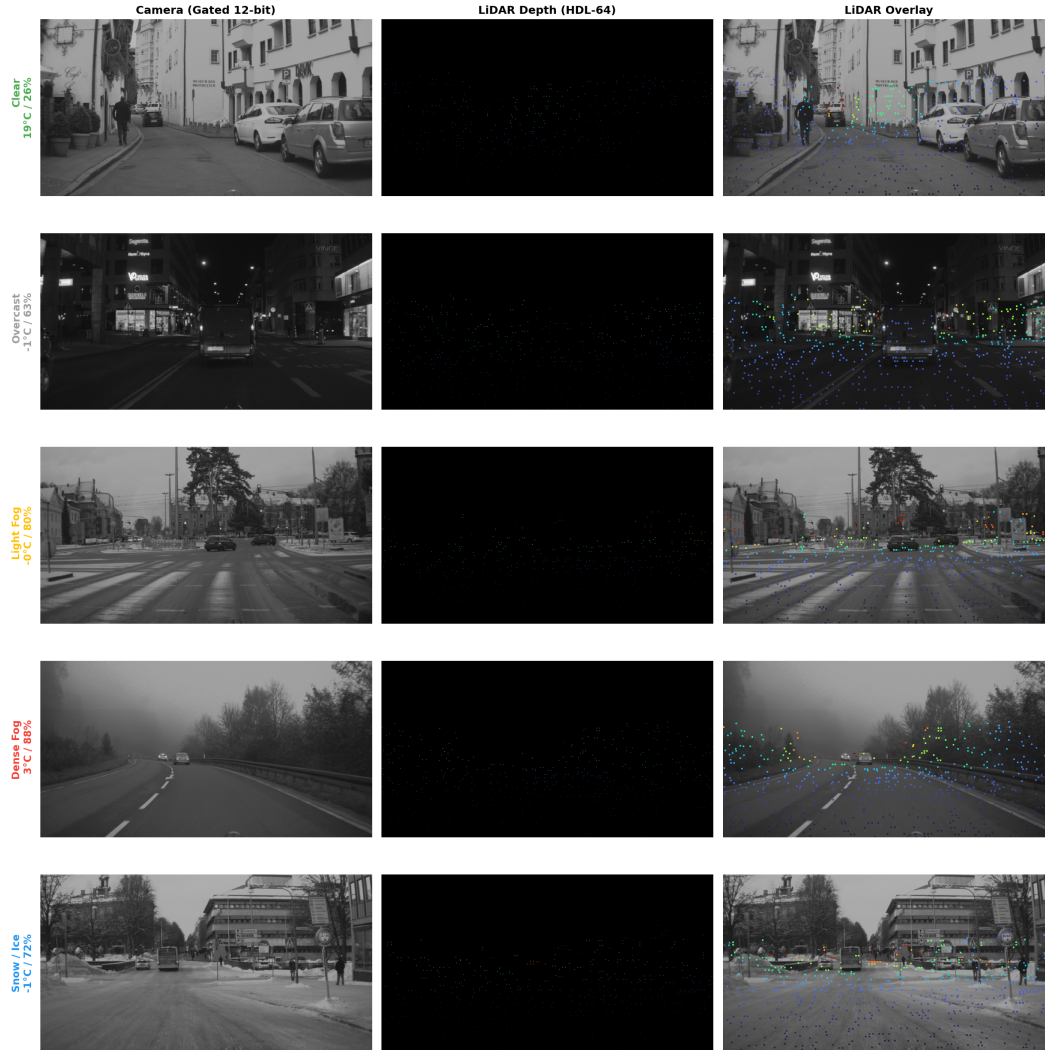


Figure 10: **STF sensor overlay per weather condition.** Sparse Velodyne HDL-64 LiDAR returns (centre, right) highlight the multi-modal perception challenge addressed by PhysDNet. Imagery from the monochrome 12-bit gated NIR camera.

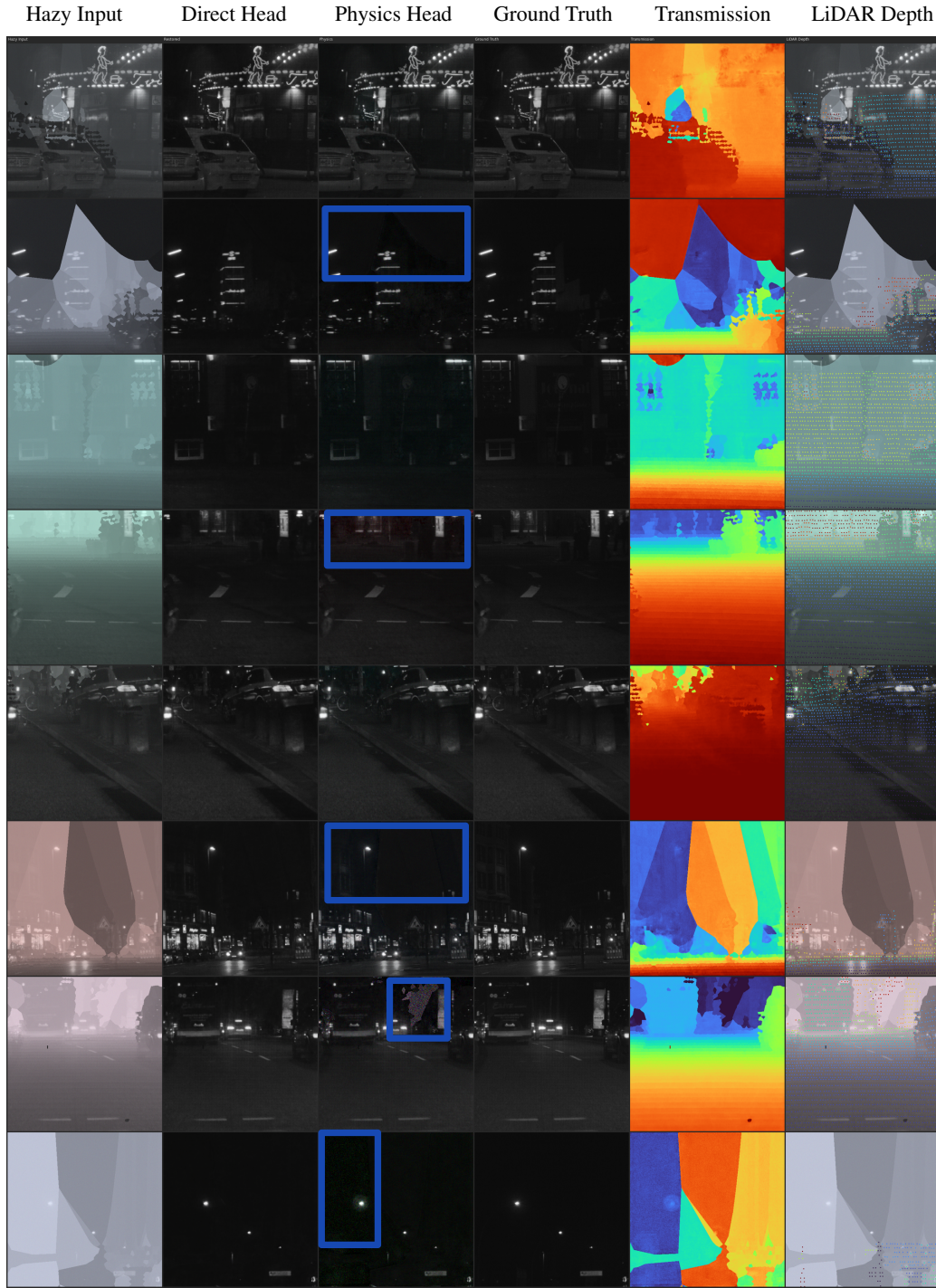


Figure 11: **Dual-head outputs on synthetic STF fog.** Blue boxes mark dense-haze regions where the Physics Head shows noise amplification; the Direct Head suppresses these artifacts by relaxing the rigid physical assumptions.

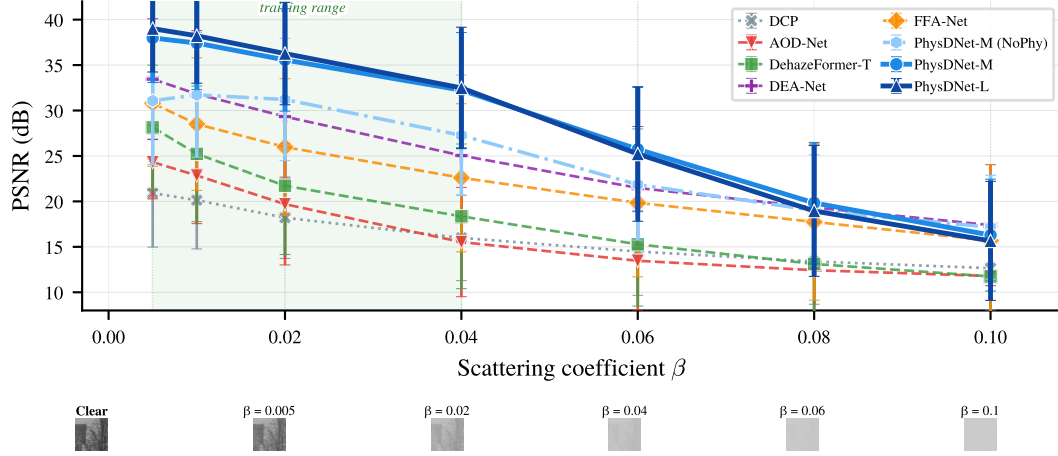


Figure 12: **PSNR vs. fog density β** . All models evaluated at 256×256 with $A=0.8$. The green band denotes the training range. PhysDNet-M maintains the highest PSNR across all densities, retaining a >6 dB advantage over the best baseline at $\beta=0.06$ and degrading gracefully outside the training distribution.

PhysDNet-M degrades smoothly beyond the training boundary, whereas baselines like FFA-Net suffer steeper drop-offs. We attribute this to the physics-constrained encoder: because the inversion formula is domain-invariant, the learned representations generalize significantly better to unseen β values.

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